

Co-optimizing Generation, Transmission, and Large-Load Siting in Expansion Planning

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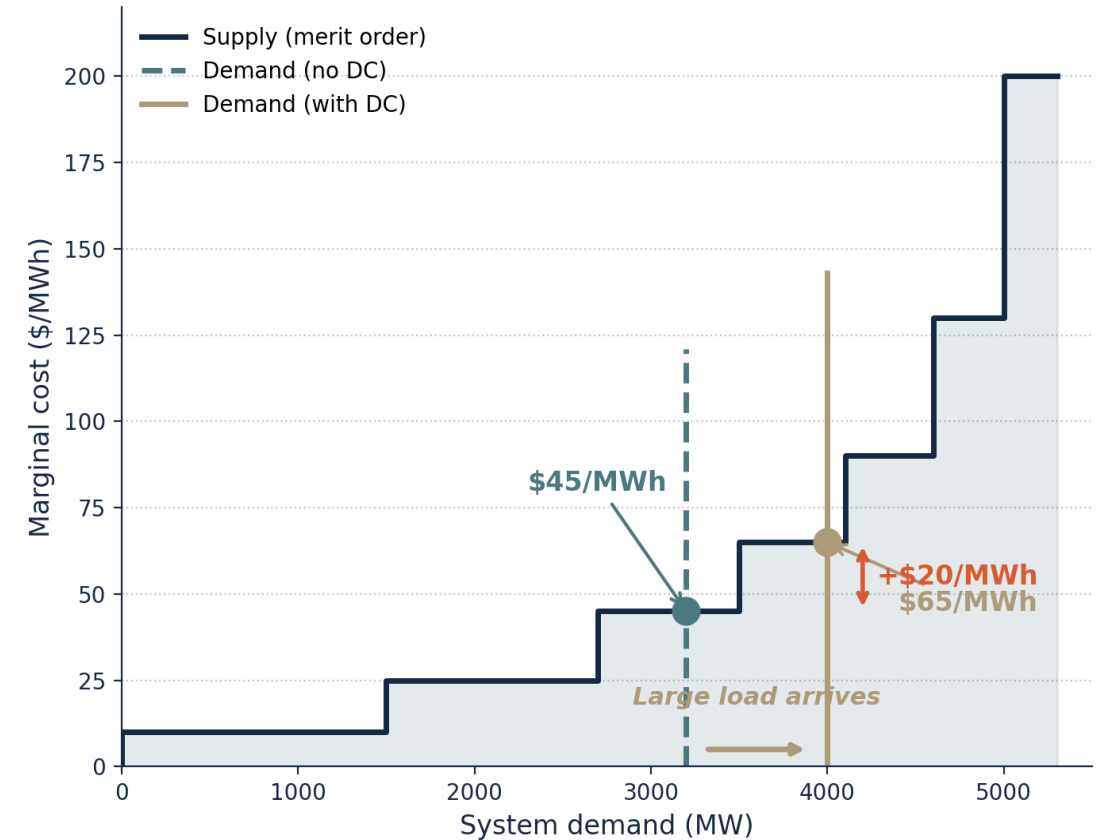
July 21th, 2026

Large Demand Blocks

When a single load is big enough to move the marginal price

- ▶ **A single large load** can change operational dispatch, which lines need reinforcement, and which fuels are economic.
- ▶ **Location matters.** A 500 MW load at a weak bus may force a transmission upgrade larger than the load's investment.
- ▶ **Profile matters.** Whether peaky or flat changes the system's capacity needs.
- ▶ **Flexibility matters.** If the load can shift, the system can absorb it without new peakers.
- ▶ **And in long-term planning,** the decision to build the load itself belongs in the optimization — **not as an exogenous input.**

Data centers are the most interesting case study right now — and the economics are genuinely different from anything else on the grid.



Data centers as a flexibility example

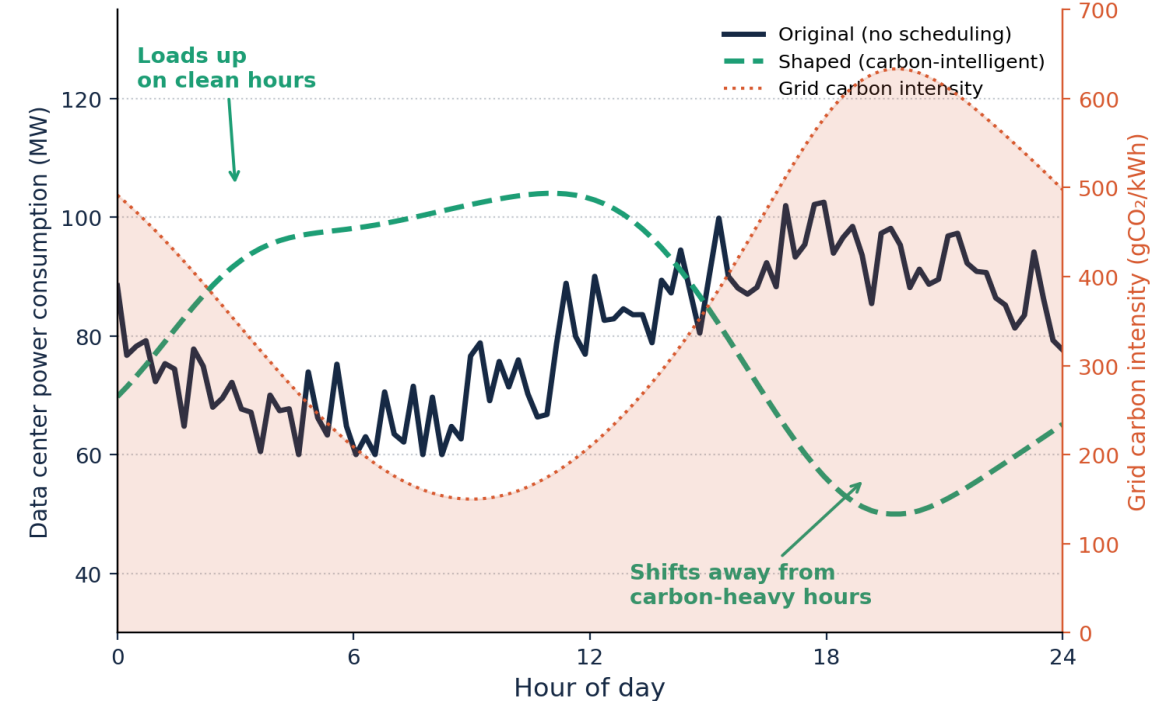
Massive new loads with built-in flexibility



Data center facilities: enormous, growing, and physically flexible

- Data center facilities are now ~40-500+ MW each — comparable to a nuclear unit, but as a single customer.
- Compute workloads can be scheduled — non-urgent jobs can be shifted across hours of the day.
- A data center can **reduce consumption in one hour and increase it in another**, keeping daily energy constant.

→ Maps cleanly onto the flexible demand in SDDP



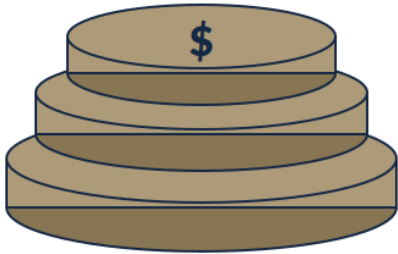
Concept based on Google Cloud Blog (2020) and Radovanovic et al., "Carbon-Aware Computing for Datacenters" (2021)

Source: Google Cloud Blog, "Using location to reduce our computing carbon footprint" (2021); Radovanovic et al., IEEE Trans. Power Syst. (2022).

Modeling data center in OptGen

Three inputs the user provides — three faces of the same investment decision

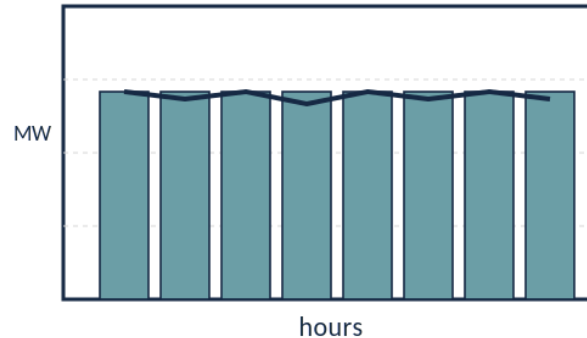
Investment
Million USD



(1) Investment cost

Construction cost in MUSD,
like any expansion candidate

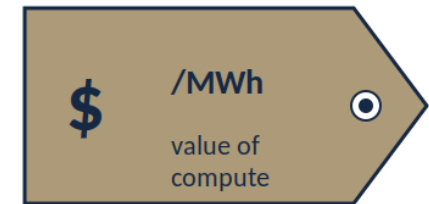
Demand profile
MW per hour



(2) Demand profile

Hourly profile and flexibility

Willingness to pay
USD/MWh



(3) Willingness to pay

Price (USD/MWh) representing
the value of compute per MWh

The willingness to pay (3) is the link between the *physical grid model* and the *data center business model*.

Data center economics in OptGen

How the optimizer decides to build, run, or skip the data center

$$\text{NetRevenue} = \sum [(P_{DC} - \pi_h) \cdot D_h^{flex}] - C_{DC}^{inv}$$

P_{DC} : value of compute (USD/MWh)

D_h^{flex} : hourly energy supplied (MWh)

π_h : marginal cost (dual, USD/MWh)

C_{DC}^{inv} : annualized investment (USD)

- ▶ **Per-hour margin:** every MWh supplied earns $(P_{DC} - \pi_h) \times D_h$ — positive when system is cheap, negative when system is stressed.
- ▶ **When demand is not supplied** (flexible mode skips a high-cost hour): the DC *shifts* the revenue to another hour
- ▶ **Build decision:** OptGen builds the DC only if **NetRevenue > 0** across the planning horizon — pure economic logic.

***What is elegant:** the DC is reasoned about with the same logic as a wind farm or any generation candidate — "invest if expected revenue beats CAPEX" — just with the sign of the energy flow reversed.*

Calibrating the price: two references

Two very different USD/MWh values

Reference 1

*What large loads actually pay for power
(revealed contract prices)*

$P_{DC} \approx \text{Contract price (revealed)}$

- ▶ Three Mile Island PPA: \$45–70/MWh
- ▶ Susquehanna nuclear: \$65/MWh
- ▶ Comanche Peak: \$90/MWh
- ▶ Crane Clean Energy Center: \$101/MWh
- ▶ Premium above zonal forwards: +\$34/MWh
- ▶ First-of-a-kind SMR projects: up to \$331/MWh

Range: ~\$60–100/MWh (firm power)

Reference 2

*What the compute is actually worth per MWh
(economic value of consumption)*

$P_{DC} \approx \text{Annual revenue} / \text{Annual energy}$

- ▶ Mid-size DC (30 MW, ~\$300M CAPEX) needs ~\$100M/yr revenue
- ▶ → implies energy value of \$400+/MWh
- ▶ AI facilities (GPU clusters): thousands/MWh
- ▶ McKinsey: DC WTP is an outlier vs. all other uses

Range: ~\$400–2,000+/MWh.

Which one goes into OptGen? → Reference 2. The investment cost being recovered is the data center, not just the power supply.

Worked numerical illustration

100 MW DC · P(DC) = \$400/MWh · C(inv) = \$50M/yr

Scenario A

Low-cost system

Avg. marginal cost: \$40/MWh

Hourly margin: $(400 - 40) \times 100 = \$36,000/\text{h}$

Annual margin: $36,000 \times 24 \times 365 \approx \$315\text{M}/\text{yr}$

Annualized investment: \$50 M/yr

Net annual revenue: +\$265 M/yr

✓ **BUILDS**

Scenario B

Stressed system

Avg. marginal cost (DC window): \$350/MWh

Hourly margin: $(400 - 350) \times 100 = \$5,000/\text{h}$

Annual margin: $5,000 \times 24 \times 365 \approx \$43.8\text{M}/\text{yr}$

Annualized investment: \$50M/yr

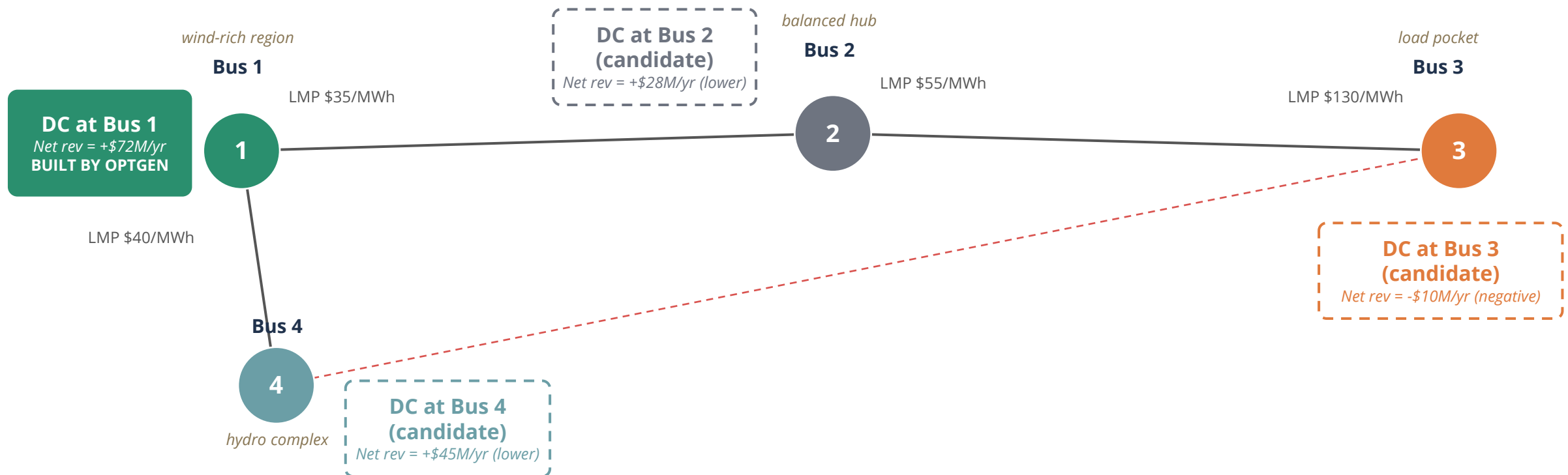
Net annual revenue: -\$6.2M/yr

✗ **DOES NOT BUILD**

The location decision — which bus?

OptGen doesn't just decide whether to build — it decides where

OptGen evaluates the same DC at every candidate bus — LMPs, transmission upgrades triggered, interaction with the rest of the expansion plan and picks the location that not only maximizes the data center's net revenue but also minimizes the system's total (inv + oper) costs



The output is not just “*build it or not*” — it’s “*build it at bus X, with this capacity and here are the transmission reinforcements that come with it, and here is how the rest of the expansion plan changes as a consequence.*”

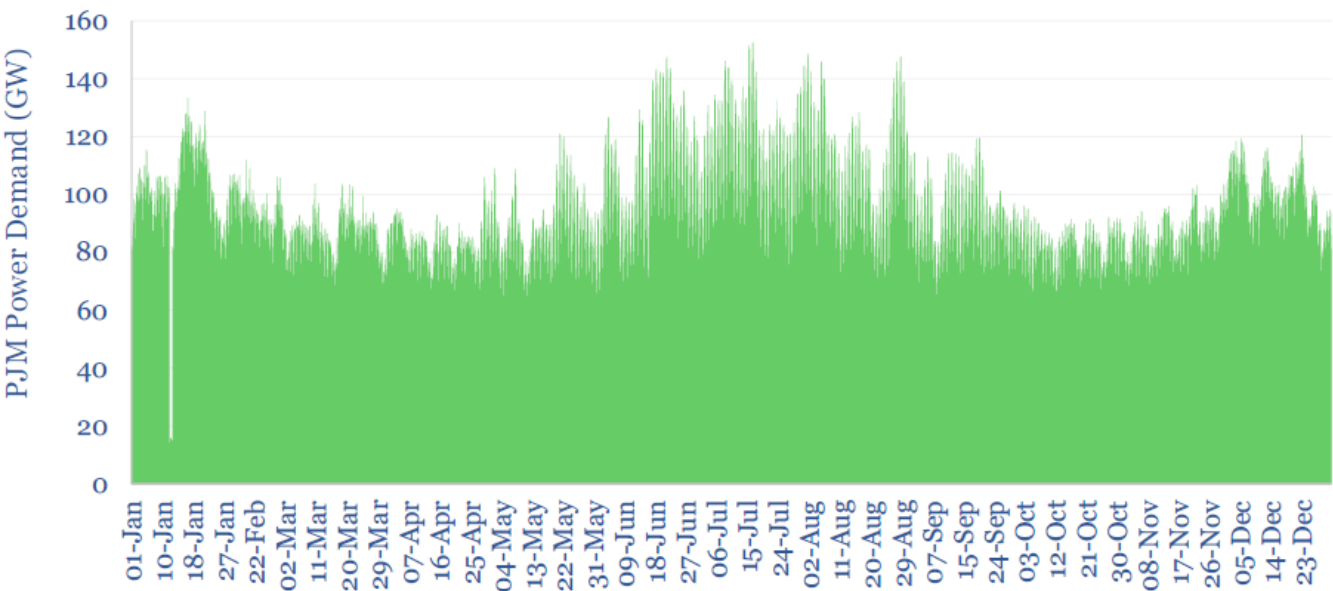


Stochastic Production Costing Adds Value to DC Studies

Deterministic screening understates cost and risk — only full stochastic simulation captures the flexibility value of data-center demand

DATA CENTER FLEXIBILITY

Thunder Said Energy, March 2026 – Flexible data center: can AI load shift? It can take 4-12 years to expand the grid and accommodate large new AI data centers. But what if more flexible data centers could be energized mostly via the pre-existing grid?



153 GW (1 hour/0.01% of all the hours)

150 GW (8 hours/0.1%)

140 GW (90 hours/1%)

It is possible to add 13 GW of new load in the system if it is willing to curtail its activity during 90 hours per year



Thunder Said Energy

30 March 2026

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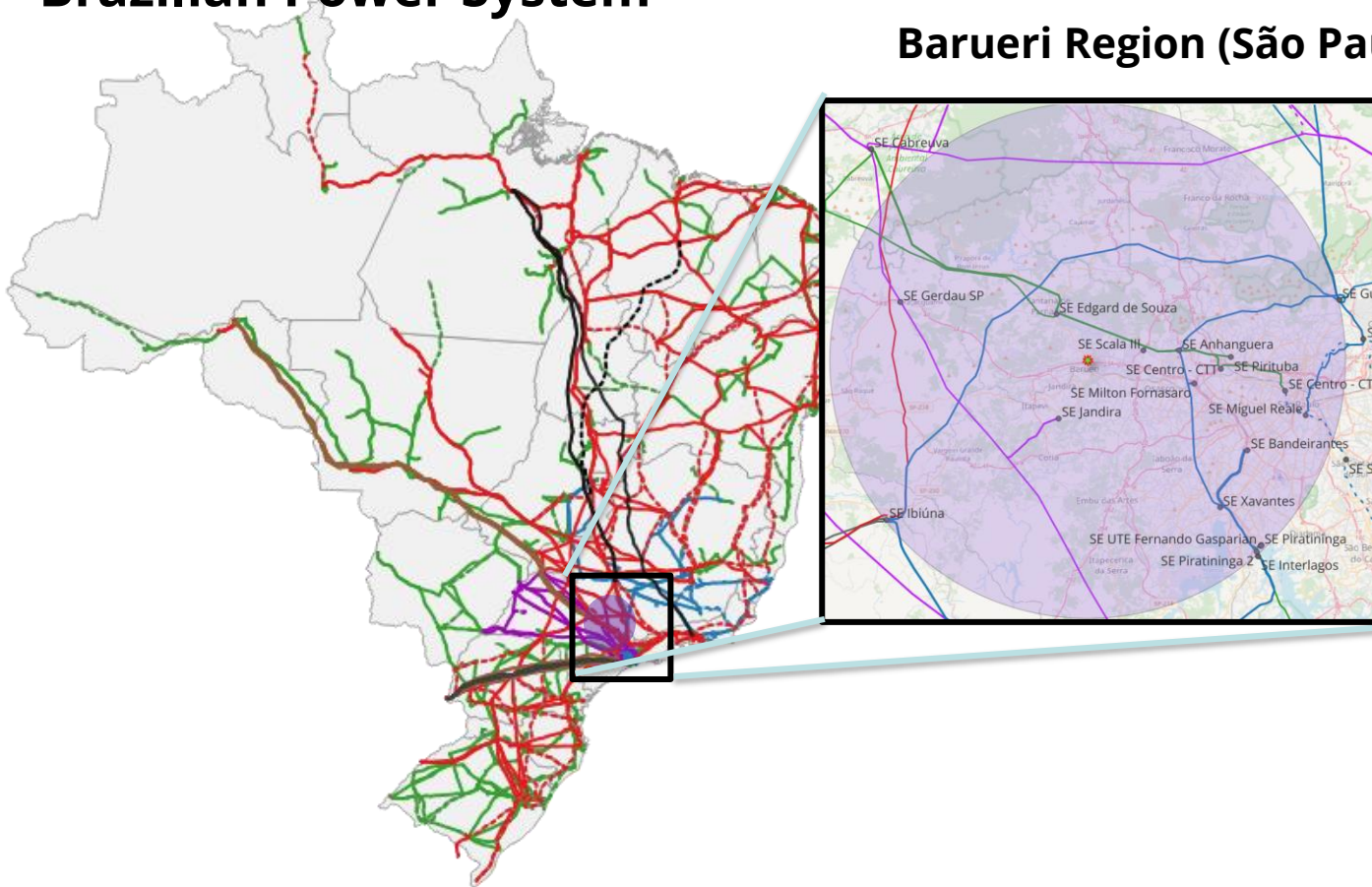
Flexible data centers: can AI load shift?

It can take 4-12 years to expand the grid and accommodate large new AI data centers. But what if more flexible data centers could be energized mostly via the pre-existing grid? This report shows how flexible AI data centers are technically feasible, economically justified, and accelerating?

STUDY CASE: BRAZIL

Brazilian Power System

Barueri Region (São Paulo)

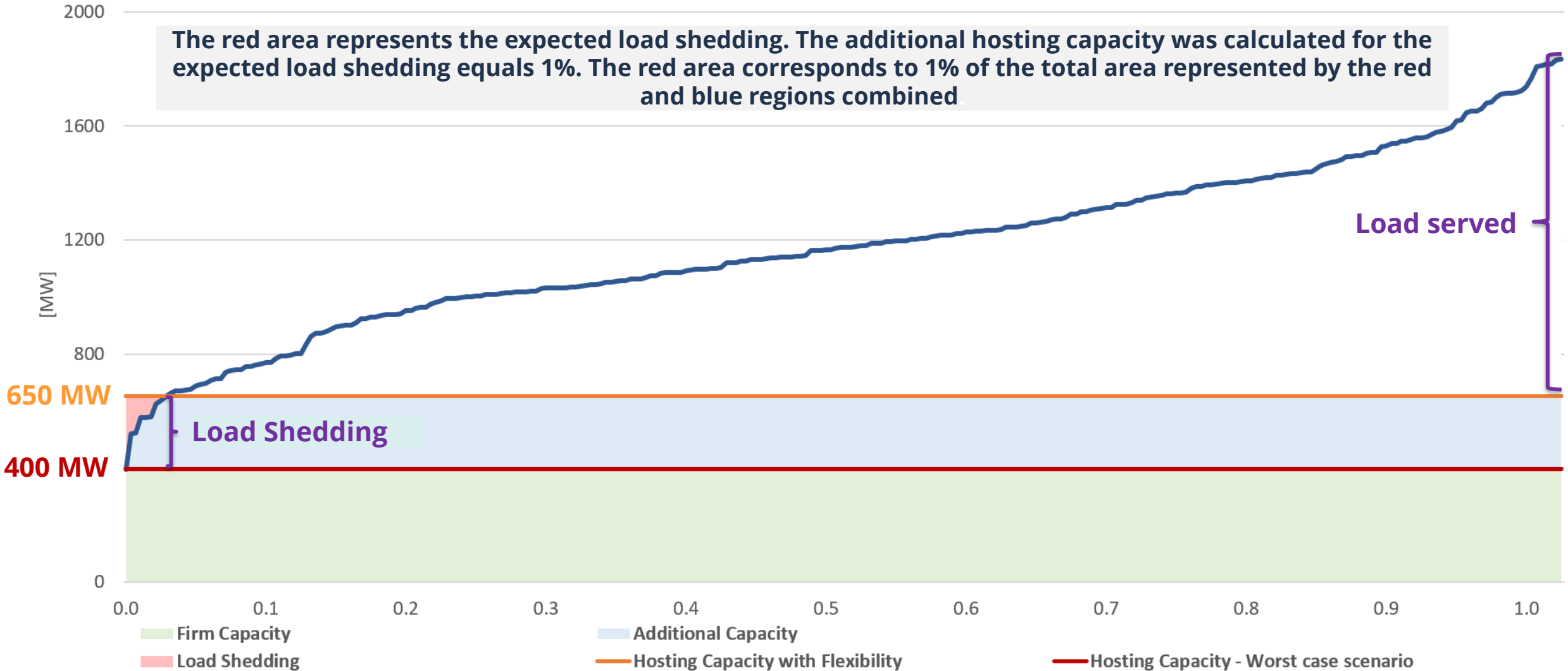


Simulation Details



2029
200 scenarios
12 months
Typical days

STUDY CASE: BRAZIL



FLEXIBILITY ASSUMPTION

How much flexibility (MW and hours/year) is required to enable an earlier connection?

Deterministic Analysis (**Conservative Assessment**)

I can only determine the available connection capacity under the worst-case scenario, which is 400 MW for the proposed case study.

Stochastic Analysis (**Flexibility value evaluated**)

Analyzing the full range of operating conditions unlocks an additional 250 MW of connection capacity, provided if the customer accepts curtailment of up to 1% of the additional load per year.

The most critical periods are February daytime hours (12:00–17:00), followed by March and April.

THANK YOU!

Any questions?

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